

50 Years of Changes—How to Brace Yourself!

SIGGRAPH 2023 Retrospective Panel

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ABSTRACT

Having lived through the 50 years of changes, the panelists attempt to put them in calibrated perspective, not merely for the sake of fond reminiscence—and fun!—but as a guide to those who face the looming future changes.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics.

KEYWORDS

history, panel

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1 INTRODUCTION

The six panelists each embrace the entire 50 years of SIGGRAPH in their careers, and are mutual friends and colleagues. Award winners all, they are leaders in art, science, education, and business. To celebrate SIGGRAPH's 50th anniversary they survey the massive changes that they have witnessed—if not caused—not only for the joy of reminiscence but also to indicate what the future of SIGGRAPH might feel like, to calibrate the awesome rate of change and to prepare attendees, including themselves, for the vaster changes looming ahead. And to convey the sense of nonstop fun they have experienced surfing the supernova computational explosion.

2 THE PANEL

Andries van Dam leads off the panel. He is a professor at Brown University. He co-authored nine books, including *Fundamentals of Interactive Computer Graphics* and three editions of *Computer*

Graphics: Principles and Practice. Fellowships: ACM, IEEE, AAAS, Computer History Museum; Memberships: SIGGRAPH Academy, NAE, American Academy of Arts & Sciences; Awards: Coons Award, SIGGRAPH Distinguished Educator Award.

Andy paints a picture of the “before times,” the decade before the first SIGGRAPH conference, when computers were just shifting from running in “batch” mode and interactive displays were exceedingly rare. Sutherland's revolutionary Sketchpad was inspiring others to work in interactive graphics. Andy describes his cofounding of ACM SIGGRAPH, a forerunner to SIGGRAPH, with some local chapters but no national presence and no conference. Interactive graphics remained a niche field that first decade. Vector graphics dominated, even as raster graphics waited in the wings.

James Blinn follows with the changes from the earliest computer graphics conferences (before SIGGRAPH) to the latest SIGGRAPH. He is best known for his computer animations at JPL of the NASA Voyager planetary fly-bys for Carl Sagan's *Cosmos* TV series, and for the telecourses *The Mechanical Universe* and *Mathematics!* He did famous work on lighting and texturing, and has collected his *IEEE CG&A* columns into three books. Fellowships: MacArthur; Memberships: NAE; Awards: Coons Award, SIGGRAPH Computer Graphics Achievement Award, Ub Iwerks Award, NASA Exceptional Service Medal.

There is an old saying in the engineering profession: cheap, fast, and good; pick any two. But computer graphics has managed to nail all three of these goals. Jim covers some of the ways we have gone from expensive, slow, and bad graphics to our current much happier state. He tells various scary stories of the bad old days, and describes the many transitions he has seen. Additionally he describes how the SIGGRAPH conference itself has evolved from a few people looking at a blackboard in a college classroom to the multimedia extravaganza that we now enjoy.

James Foley is professor emeritus at Georgia Tech. He co-authored computer graphics texts with ~400,000 copies in print. From 1996-99 he directed Mitsubishi Electric's US Research Labs; from 2001-05 he chaired the Computing Research Association. He was twice SIGGRAPH vice-chair, 1974-76 and 2008-11. Fellowships: AAAS, ACM, IEEE; Memberships: NAE; Awards: SIGGRAPH and SIGCHI Lifetime Achievement Awards.

Having attended 49 of the 50 SIGGRAPH conferences, Jim “walks us down memory lane” showing how the conference reflected changes in what we mean by “Computer Graphics and Interactive Techniques.” Its papers often foreshadowed future hardware

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and software exhibits, and its courses presented both basic concepts and recent research results. Its early exhibits were all about vector graphics and (mostly) 2D design applications, often implemented with graphics terminals attached to mainframes, but increasingly, after a series of hardware “wars,” movie production software powered by GPUs came to dominate. But, through all these changes, there has been a constant: computer graphics is about ways to model, interact with, and visualize real and imaginary constructs.

Alvy Ray Smith (moderator) cofounded Pixar and Altamira Software, was first Director of Computer Graphics at Lucasfilm, and first Graphics Fellow at Microsoft. He directed two early mileposts—Genesis Demo in *Star Trek II: The Wrath of Khan* and *The Adventures of André & Wally B.*—wrote the first RGB paint program, and co-invented the alpha channel. His book, *A Biography of the Pixel* (2021), includes an extensive history of computer graphics. Fellowships: AAAS; Memberships: SIGGRAPH Academy, NAE; Awards: SIGGRAPH Computer Graphics Achievement Award, two technical Academy Awards.

Based on a decade of thinking and research for his book, Alvy puts things in historical and meaningful perspective. He recasts Moore’s Law into an intuitive form that calibrates its awesome influence—and explains why we cannot know the future. He propounds a notion, the Central Dogma, that has been with us for over 50 years, but unarticulated, which helps define and unify the many parts of modern computer graphics. And he suggests that we have just departed the code-and-debug era—perhaps leading to garage-band Pixars.

Turner Whitted is best known for coining the term “global illumination” and implementing it using recursive ray tracing. His

broad-ranging research career includes terms at Bell Labs, Microsoft Research, and Nvidia. In addition he cofounded Numerical Design Limited, and served as faculty at the University of North Carolina and North Carolina State University. Fellowships: IEEE, ACM; Memberships: NAE; Awards: Coons Award.

A platform for computer graphics contains, at least, a computer and a display. For performance and convenience we benefit from an intermediate computational element between the computer and the display as well as software layers. The evolution of graphics platforms since the early 1970s involves advances in silicon design, manufacturing, algorithms, and graphics software interfaces for speed, realism, and availability. Turner highlights some of the key developments along this trail that has given us our dream.

Donald Greenberg closes the panel with a message to SIGGRAPH. Don, a professor at Cornell University, has been researching and teaching since 1965. He has taught computer graphics, CAD, computer animation, and technology strategy in different Cornell schools. He and his formidable students have made Cornell one of the foremost centers of computer graphics research.

“For more than a half-century I have been teaching computer graphics and using the immense power of pictorial images in many disciplines: training astronauts, repairing aortic aneurysms, designing buildings, studying ivory-billed woodpeckers, animation, and special effects. SIGGRAPH has been my academic home, and I am forever grateful. But now, with the exponential growth of computer power, the continuing accuracy of measurement technology, and immense proliferation of data, is it not time for a change of direction? Can we not modify our goals from generating exquisite images to designing a better world and improving the performance of our products and designs?”